

Part I. Shorter problems (9 points each). Give brief but to the point justification.

1. Let A and B be two 5×5 matrices over $\mathbb{Q}$ that both have the characteristic polynomial $x^5 + 3x + 3$. Prove that A and B are similar.

Solution. By the Eisenstein criterion applied to $p = 3$, $f = x^5 + 3x + 3$ is irreducible over $\mathbb{Q}$. Hence, the invariant factors of both A and B are $1, 1, 1, 1, f$.

2. Compute the greatest common divisor of $5 + i$ and 13 in the ring $\mathbb{Z}[i]$ of Gaussian numbers.

Solution. One has $13 = (3 + 2i)(3 - 2i)$, where both $3 + 2i$ and $3 - 2i$ are irreducible. We have $(5 + i)/(3 - 2i) = (5 + i)(3 + 2i)/13 = 1 + i$, which is irreducible. So the gcd is $3 - 2i$. Alternatively, one can apply the Euclidean algorithm.

3. Let R be a ring, and let M be a Noetherian left R -module. Let $f : M \rightarrow M$ be a surjective homomorphism. Prove that f is an isomorphism.

Solution. The chain $\ker(f) \subset \ker(f^2) \subset \ker(f^3) \subset \dots$ stabilizes. Assume $\ker(f^n) = \ker(f^{n+1})$. Now the surjection $\ker(f^{n+1}) \rightarrow \ker(f) : x \mapsto f^n(x)$ with kernel $\ker(f^n)$ shows that $\ker(f) = 0$, so f is also injective.

4. Prove that if I and J are ideals in $\mathbb{C}[x_1, \dots, x_n]$ such that IJ is a radical ideal, then $IJ = I \cap J$.

Solution. One has $V(IJ) = V(I) \cup V(J) = V(I \cap J)$. Hence, by Hilbert's Nullstellensatz, $I \cap J$ is contained in the radical of IJ , so $I \cap J = IJ$.

5. Let k be a field, and let R be a simple non-commutative k -algebra such that $\dim_k R = 4$. Prove that the center of R is k .

Solution. One has $R = M_n(D)$, where D is a division ring over k . If $n = 2$ then $D = k$, and the center of R is k . Assume now $n = 1$, and let $C \subset D$ be the center. Since D is noncommutative, we only have to exclude the case $\dim_k C = 2$. But in this case $\dim_C D = 2$, so D is generated as C -algebra by one element, hence commutative, a contradiction.

6. Determine the automorphism group of $C_2 \oplus C_4$.

Solution. This is D_8 . Indeed, let e_1, e_2 be generators of C_2 and C_4 , so that $2e_1 = 4e_2 = 0$. There are two cyclic subgroups of order 4 in $C_2 \oplus C_4$: $\langle e_2 \rangle$ and $\langle e_2 + e_1 \rangle$. Let α be an automorphism that preserves each of these two subgroups. Then $\alpha(e_2) = \pm e_2$. Suppose $\alpha(e_2) = e_2$. Then $\alpha(e_2 + e_1)$ is either $e_2 + e_1$ or $-e_2 + e_1$, so $\alpha(e_1)$ is either e_1 or $e_1 + 2e_2$. Let σ denote the automorphism sending e_2 to e_2 and e_1 to $e_1 + 2e_2$. Then $\sigma^2 = \text{id}$ and the subgroup of automorphisms preserving both cyclic subgroups of order 4 is $\{\pm \text{id}, \pm \sigma\} \simeq C_2 \times C_2$ (where $-\text{id}$ sends e_1 to e_1 and e_2 to $-e_2$). Now let τ denote the automorphism sending e_2 to $e_2 + e_1$ and e_1 to e_1 . Then $\tau^2 = \text{id}$, and $\tau\sigma\tau = -\sigma$.

The isomorphism with D_8 sends σ and τ to a pair of generating reflections such that $\sigma\tau$ is an element of order 4.

II. Longer problems (12 points each). Do any *four* of the following problems (only 4 problems with highest scores will be counted).

1. Let R be the ring of upper-triangular $n \times n$ -matrices over $\mathbb{Z}/9\mathbb{Z}$. What is the Jacobson radical of R ? Classify irreducible R -modules.

Solution. Let J denote the subset of matrices in R with diagonal entries in $3\mathbb{Z}/9\mathbb{Z}$. It is easy to see that J is a two-sided ideal, whose every element is nilpotent, hence, $J \subset J(R)$. Furthermore, R/J is isomorphic to the direct sum of n copies of $\mathbb{Z}/3\mathbb{Z}$, which is a semisimple ring. Hence, $J = J(R)$. Irreducible R -modules are in bijection with irreducible R/J -modules, so there is one for each copy of $\mathbb{Z}/3\mathbb{Z}$. So there are n pairwise non-isomorphic irreducible R -modules.

2. a) For an element g of a finite group there is a unique decomposition $g = xy = yx$ where the order of x is a power of 2 and the order of y is odd.

Solution. Let $n = 2^r m$ be the order of g , where m is odd. Since m and 2^r are relatively prime we can write $1 = am + b2^r$. Now we set $x = g^{am}$, $y = g^{b2^r}$. Conversely, assume $g = xy = yx$, where $x^{2^{r'}} = 1$ and $y^{m'} = 1$. Then $x^{m'} = g^{m'}$, $y^{2^{r'}} = g^{2^{r'}}$, which implies that x and y are powers of g . If $x = g^c$, $y = g^d$ then c is divisible by m , d is divisible by 2^r and $c + d \equiv 1 \pmod{mn}$, so we recover the above solution.

(b) The map $g \mapsto x$ (where x is as in part (a)) is an endomorphism θ of forgetful functor **FiniteGroups** $\rightarrow$ **Sets**.

Solution. This amounts to the statement that if $\phi : G \rightarrow G'$ is a homomorphism of groups, and $g = xy$ is the decomposition of $g \in G$ as in (a), then $\phi(g) = \phi(x)\phi(y)$ is a similar decomposition of $\phi(g) \in G'$.

(c) The endomorphism θ does not lift to the category **Groups**: there is no endomorphism ψ of the forgetful functor **Groups** $\rightarrow$ **Sets** such that ψ restricted to **FiniteGroups** is θ .

Solution. For every $G = \mathbb{Z}/m$, where m is odd, $\theta_{\mathbb{Z}/m}$ sends 1 to 0. If θ extends to the category of all groups, then this would imply that $\theta_{\mathbb{Z}}$ sends 1 to 0. But this contradicts to the fact that $\theta_{\mathbb{Z}/2}$ sends 1 to 1.

3. Let V be an irreducible complex representation of S_n and let $V' = V \otimes \text{sign}$, where sign is the sign representation of S_n . Prove that the following are equivalent:

- (i) V' is equivalent to V as S_n -representation;
- (ii) The restriction of V to A_n is reducible;
- (iii) $\chi_V(\sigma) = 0$ for any odd permutation σ , where χ_V is the character of V .

Solution. Equivalence of (i) and (iii) is established by considering the equality of characters of V and $V \otimes \text{sign}$. Let us show the equivalence of (i) and (ii). Let Res and Ind denote the restriction and induction functors between representations of S_n and A_n . By Frobenius reciprocity, $\text{Hom}_{A_n}(\text{Res}(V), \text{Res}(V)) \simeq \text{Hom}_{S_n}(\text{Ind}(\text{Res}(V)), V)$. On the

other hand, one has $\text{Ind}(\text{Res}(V)) \simeq V \oplus V \otimes \text{sign}$. Since V is irreducible, (i) is equivalent to the condition $\dim \text{Hom}_{S_n}(\text{Ind}(\text{Res}(V)), V) > 1$, i.e., $\dim \text{Hom}_{A_n}(\text{Res}(V), \text{Res}(V)) > 1$, which is equivalent to $\text{Res}(V)$ being reducible.

4. Prove that any group of order $255 = 3 \cdot 5 \cdot 17$ is cyclic.

Solution. By Sylow's theorems, the number of Sylow 17-subgroups is $\equiv 1 \pmod{17}$ and divides 15. Hence, there is a unique Sylow 17-group $C_{17} \simeq P \subset G$ and it is normal. The quotient G/P is a cyclic group of order 15. The action of G/P on P by conjugation is given by a homomorphism $G/P \rightarrow \text{Aut}(C_{17}) \simeq C_{16}$. Hence, this homomorphism is trivial, so P is central. Since G/P is cyclic, this implies that G is commutative. Hence, by classification of finite abelian groups, $G \simeq C_3 \times C_5 \times C_{17} \simeq C_{255}$.

5. Find all intermediate subfields F between $\mathbb{Q}$ and $K = \mathbb{Q}(\zeta)$, where $\zeta = \zeta_7$ is a primitive 7th root of unity. Express each such subfield as $F = \mathbb{Q}(\alpha)$ for some $\alpha \in K$.

Solution. The Galois group G of $K/\mathbb{Q}$ is isomorphic to $\mathbb{Z}_7^*$, which is cyclic of order 6, generated by -2 . The corresponding generator σ of G sends ζ to ζ^{-2} . The intermediate subfields correspond to subgroups of G . Consider first the subgroup $\langle \sigma^3 \rangle$ of index 3. The corresponding cubic subextension $F_1 \subset K$ consists of invariants of σ^3 , which sends ζ to ζ^{-1} (so this is just the complex conjugation). Since $1, \zeta, \dots, \zeta^6$ is a basis of K over $\mathbb{Q}$, F_1 is spanned over $\mathbb{Q}$ by the elements

$$1, \zeta + \zeta^{-1}, \zeta^2 + \zeta^{-2}, \zeta^3 + \zeta^{-3}.$$

Now we observe that $(\zeta + \zeta^{-1})^2 = \zeta^2 + \zeta^{-2} + 2$, while $(\zeta + \zeta^{-1})^3 = \zeta^3 + \zeta^{-3} + 3(\zeta + \zeta^{-1})$. Hence $F_1 = \mathbb{Q}(\zeta + \zeta^{-1})$. Next, consider the subgroup $\langle \sigma^2 \rangle$ of index 2. The corresponding quadratic subextension $F_2 \subset K$ is generated over $\mathbb{Q}$ by an element not in $\mathbb{Q}$, so $F_2 = \mathbb{Q}(\zeta + \zeta^2 + \zeta^4)$. All intermediate subfields are $\mathbb{Q}$, F_1 , F_2 and K .